

Latch risk

Combinational logic should depend only on current inputs, not on what the output was last cycle

Complete versus incomplete paths

- A continuous assign is purely combinational, there is no path where Y fails to update
- In always at-star
- If you write only if open-paren S close-paren Y equals D1 with no else
- A full if-else or a case with all labels covered, or a default, gives every path a value
- At-star fixes sensitivity; it does not fix missing branches

Browser lab

Digital Design and Verification Platform

HomeTools

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INTERACTIVE TOOL

Combo style / latch-risk checker

Compare continuous assign with always @(*) / case styles for the same function.
Incomplete branches hold the previous value — inferred latch. Starter: a 2:1 mux written three ways.

Starter example: 2:1 mux as assign (combo OK). Switch to “always incomplete” to see latch risk.

Challenge 0 / 22 cleared

Quiz: what is a latch?: In this lab, an inferred latch means...

☐ Y keeps its previous value when not assigned in @(*)

☐ a flip-flop with a clock

☐ only a NAND-gate SR cell in the library

☐ an always_ff mistake only

Show hint

Check

Next

Load challenge setup

Idle

Quiz: what is a latch?

Quiz: assign

Quiz: missing else

Quiz: case default

Quiz: partial case

Quiz: @(*)

Pick: mux latch

Pick: mux safe

Fix: enable else

Fix: case default

See: priority latch

Fix: priority else

Pick: enable latch

Pick: partial decode

Quiz: meant a flop?

Quiz: complete if

See: mux incomplete

See: mux assign OK

Quiz: full_case

Quiz: same function

Pick: priority safe

See: decode assign

Style explorer

Load starter example

Latch risk lab starter

Real Verilog practice

```
wsl - module07-latch-risk/examples/navigation

# Track A - real Unix
# real Verilog session (Track A)

$ cat examples/mux_styles/mux_styles.v
// mux_styles.v - complete assign vs incomplete always
module mux_good(
    input S, D0, D1,
    output Y
);
    assign Y = S ? D1 : D0;
endmodule

module mux_latch_risk(
    input S, D0, D1,
    output reg Y
);
    always @(*) begin
        if (S) Y = D1;
        // no else - Y not assigned when S=0 → latch risk
    end
endmodule

$ iverilog -t null examples/mux_styles/mux_styles.v
(syntax check passed)
```

Real Verilog — mux with and without latch risk

Pitfalls to watch

- Do not assume at-star means latch-free, it only builds the sensitivity list
- In coursework you usually do not
- Partial case without default is the same class of bug for wider selects
- Prefer assign or always_comb with every path assigning outputs when you mean pure

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and identify which mux style infers a latch
- On real Verilog, write or fix an always block so every branch assigns the output
- When you are ready